Andrew **Charabin**

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Work Experience

Loblaw Digital — Toronto, Ontario

Engineering Manager – Data Science & AI — *Mar. 2025 – Present*

- Recruited and led a team of 5 engineers accountable for machine learning & AI solutions spanning programmatic ad serving to campaign analytics at Advance, Loblaw's media platform.
- Enabled a new display advertising product on Loblaw's web properties by developing an impression forecasting and booking service, generating $15M in first-year revenue.
- Launched an agentic service that creates and manages advertising campaigns through a chat interface, driving a 10% sales lift by reducing friction in the sales funnel.
- Increased sponsored search revenue by 15%+ by deploying a real-time click through rate (CTR) model to optimize ad placements.
- Designed a bid valuation service to recommend bid prices for ad opportunities; demonstrating a 20% increase in ROAS while maintaining media revenue.
- Led development of a multi-agent service combining RAG-powered FAQ and analytics agents with persistent chat-history and context-aware authentication.
- Built an evaluation framework for a custom agentic workflow using recovery, tool success, and hallucination metrics; introduced LLM-as-a-judge evaluation and scaled a human-derived golden dataset; guiding refinements that achieved 100% task success rate.

Greenstone Financial Services — Toronto, Ontario

Analytics Lead — *May 2024 – Mar. 2025*

- Led the Canadian analytics portfolio including business intelligence, marketing mix modelling, and agent lead allocation; recruited and managed a Data Analyst.
- Built a Bayesian marketing mix model to quantify channel-level incremental contribution, giving Sales and Marketing leadership a data-driven framework for budget allocation.

ADJ Industries — London, Ontario

Data Manager — *Jun. 2023 – May 2024*

- Enabled a business intelligence capability by implementing an on-premises MySQL database, saving $50K+ annually in administration costs by centralizing data collection of payroll and customer-order data.

Voices — London, Ontario

Senior Manager, Data Science & Analytics — *Oct. 2021 – Jun. 2023*

- Instituted an organization-wide data science function by recruiting, onboarding, and mentoring 2 Data Scientists and 1 Data Analyst, creating a competitive edge for a global voice marketplace.
- Led a technology team responsible for centralized metrics, insights, and machine learning solutions across matching, pricing, and search to improve platform retention and liquidity.
- Reduced the time to market for new business intelligence reports and ML solutions by ~50% by leading the development of a Redshift data warehouse and ELT platform with modernized deployment, rollback, and monitoring.
- Increased job hire rate by 5 points by replacing rules-based matching with an XGBoost model that ranked voice talent by likelihood of hire.
- Contributed to a 3-point improvement in annual client retention by building a pricing recommendation system using SageMaker models to predict job price and response volume.

Manager, Analytics *Sep. 2020 - Oct. 2021*

- Established a centralized analytics capability and common picture of company performance through Tableau core metrics and inferential and causal models, supporting product and marketing strategy during 30%+ YoY growth.

Team Lead, Analytics *Aug. 2019 - Sep. 2020*

- Developed and standardized reporting and insights around platform liquidity metrics including churn, freelancer earnings, and spend retention, informing strategic discussion among leadership.

Senior Financial Analyst *Apr. 2018 – Jul. 2019*

- Developed sales forecasts to guide financial targets and designed and rolled out sales comp. plans.

Skills

AI & Engineering Python, SQL, agentic AI, LLMs, RAG, evaluation

Machine Learning Regression, classification, recommender systems, ranking, forecasting

Data Science Experimentation, causal inference, marketing mix modelling

Analytics Product & marketing, metrics, insights, visualization, data storytelling

Cloud Google (Kubernetes, Argo CD, Vertex AI, BigTable), AWS (SageMaker, Lambda)

Leadership Team building, collaboration, structuring complexity, continuous improvement

Education

Ivey School of Business London, Ontario

Master of Science in Management (MSc), Business Analytics *Jan. 2017 – Apr. 2018*

Honors Business Administration (HBA) *Sep. 2013 – Apr. 2017*

- Dean's Honor List 2014-2015 | Varsity Tennis Player, 2013 – 2018

Certifications

LangChain Academy

Certified Agent Engineer *Oct. 2026*

University of Santa Cruz

Bayesian Statistics Specialization *Apr. 2024*

Python Institute

Certified Associate Python Programmer *Feb. 2024*

DeepLearning.AI

Natural Language Processing Specialization *Jan. 2024*

University of Pennsylvania

A Crash Course in Causality: Inferring Causal Effects from Observational Data *Aug. 2023*

AWS Training & Certification

Certified Machine Learning Specialty *Dec. 2022*

Articles

Goals, Principles & Constraints *Mar. 2023*

A Framework for Organizational Leadership

CI/CD for Multi-Model Endpoints in AWS *Jun. 2023*

A simple, flexible alternative for sustainable ML solutions

Projects

Technical project portfolio available at acharabin.github.io.